Continuum Two-Soliton $O(\hbar^0)$ Phase in the Lattice-Matched Scheme: Bad-Equation Small-Amplitude Corner, Boundary Determinant Analysis, and Symmetric/Antisymmetric Separation

Continuum Two-Body Package (`twobody_cont`) — Research Note for Gustav Delius

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Abstract

We compute and audit-harden the continuum one-loop ($O(\hbar_B^0)$) two-soliton scattering phase of the **bad** Boussinesq equation at **fixed physical momenta** in the **lattice-matched renormalisation scheme** ($\Delta\mathcal{H} = -\frac{3\hbar_B}{2\pi}u^2$), in the small-amplitude corner ($\beta_1 = \varepsilon > \beta_2 = r\varepsilon$, $0 < r < 1$) and at finite amplitudes, in both the overtaking and head-on channels **(c)**. First, we show that while all implementations of $\Delta\mathcal{H}$ yield the same one-body fixed-momentum dispersion coefficient $C_{\text{matched}} = \frac{1}{2\pi} = C_T$ **(c)**, the two-body Faddeev–Korepin frame shift $\delta_{\text{renorm}} = \Delta X_1'\Delta P_1 + \Delta X_2'\Delta P_2 = H_{12}[\Delta P] + D_{12}[\Delta P]$ depends on the one-soliton momentum shift ΔP_j alone and splits into three distinct prescriptions **(c)**: (i) using the lattice-matched canonical collective momentum $p_{\text{can,matched}} = \frac{\mathrm{d}L_{1,\text{matched}}}{\mathrm{d}v} = -\frac{\epsilon}{\pi}\beta + \frac{10\epsilon}{9\pi}\beta^3 + O(\beta^5)$ eliminates all $\ln\varepsilon$ infrared divergences, drives the Hirota shift to zero ($H_{12} = O(\varepsilon^2)$), and reduces the scale-invariant Darboux shifts by a factor of 4 to $D_{\text{ov}}[p_{\text{can,matched}}](r) = -\frac{2}{\pi}(r^{-1} - r)$ and $D_{\text{ho}}[p_{\text{can,matched}}](r) = +\frac{2}{\pi}(r^{-1} + r)$ **(c)**; (ii) using the fixed-profile mode-sum momentum p_{mode} leaves $H_{\text{ov}}[p_{\text{mode}}] = \frac{2}{\pi}\ln r \ln\frac{1-r}{1+r}$ and retains a logarithmic $\pm\frac{2}{\pi}(r^{-1} \mp r)\ln\varepsilon$ divergence in $D_{12}[p_{\text{mode}}]$ **(c)**; (iii) implementing $\Delta\mathcal{H}$ by direct on-shell sound-speed rescaling $c_s = \sqrt{1 + 2\hbar_B\lambda}$ yields $\Delta_{\text{ct}}^{\text{sound}}\delta = \lambda(3\delta_0 + 4P_{0,1}\Delta X_1' + 4P_{0,2}\Delta X_2') = O(\varepsilon^2) \to 0$ in both channels **(c)**. Second, addressing open problem #2, we compute the exact finite-R configuration-space kinetic norm of the time-modulated bad-branch soliton, prove that any local analytic counterterm of order $O(\sinh^5\beta)$ vanishes in C_{matched} and in the $O(1)$ two-body frame shifts, and state the three quantities that remain undetermined on the continuum side **(c)/(d)**. Third, we separate the symmetric and antisymmetric parts under particle exchange: in the directional Bethe/Darboux frame of the Toda comparison, the lattice-matched continuum prediction for the large-Y mixed second difference (rectangle) is $\lim_{Y\to\infty}\mathcal{D}_0(Y) = -2C_{\text{matched}}(15\sqrt{2} - 20) = -\frac{15\sqrt{2}-20}{\pi} \approx -0.386175$, matching the Toda extrapolation $\mathcal{D}_0(32) = -0.39575 \pm 0.0217$ within 0.0096 (whereas the bare continued value $C_{\text{ac}} = \frac{2}{\pi}$ gives -1.544699, off by a factor of 4) **(c)**. All continuum phase components are tabulated at 55-digit precision in `twobody_cont_grid.json`.

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1 Conventions, Bad/Toda Dictionary, and Scope

We follow the ground rules of `AGENTS.md` and the audit hierarchy of `README.md` and `phase2/report.tex`: every nontrivial claim is labelled **(a)** (proved in the literature), **(b)** (conjectured or heuristic in the literature), **(c)** (derived and script-verified in this project), or **(d)** (a guess or unestablished identification), and every **(c)** formula is verified by one of the three standalone scripts

`twobody_cont_phase.py`, `twobody_cont_det_antisym.py`, `twobody_cont_grid.py`,

each ending with `ALL CHECKS PASSED`.

Good and bad Boussinesq conventions. In the repository normalisation (`AGENTS.md`, `phase1/review.tex`), the **good** Boussinesq equation is $u_{tt} = u_{xx} - u_{xxxx} - 3(u^2)_{xx}$ with potential $u = \phi_x = 2(\ln\tau)_{xx}$ and symplectic form $\Omega = \int \mathrm{d}\pi \wedge \mathrm{d}\phi$ ($\pi = \phi_t$). The **bad** Boussinesq equation, obtained by analytic continuation $\alpha = -i\beta$ ($\beta > 0$) and curated in `phase2/toda_route.tex` and `phase2/onebody_scheme.tex`, is

$$u_{tt} = u_{xx} + u_{xxxx} + 3(u^2)_{xx}, \qquad u = \phi_x = 2(\ln\tau)_{xx}, \tag{1}$$

with supersonic one-soliton parameters ($\epsilon \in \{\pm 1\}$):

$$\begin{aligned} u_0(x,t) &= -\frac{k^2}{2}\operatorname{sech}^2\frac{k(x-X-vt)}{2}, \qquad \Phi(\xi;v,C) = C - k\tanh\frac{k\xi}{2}, \\ k &= \sinh\beta, \qquad v = \epsilon\cosh\beta, \qquad \ell = \epsilon\cosh 3\beta, \qquad M = \int_{-\infty}^{\infty}(-u_0)\,\mathrm{d}x = 2\sinh\beta, \\ \Pi &= -2\epsilon\sinh\beta\cosh\beta, \qquad P_0 = \frac{2}{3}\epsilon\sinh^3\beta\cosh\beta, \qquad E_0 = \frac{2}{3}\sinh^3\beta + \frac{8}{15}\sinh^5\beta, \end{aligned} \tag{2}$$

and classical collective Lagrangian $L_0(\beta) = vP_0 - E_0 = -\frac{2}{15}\sinh^5\beta$, which satisfies $\frac{\mathrm{d}L_0}{\mathrm{d}v} = P_0$ and $\frac{\mathrm{d}P_0}{\mathrm{d}\beta} = \frac{2}{3}\epsilon\sinh\beta\sinh 3\beta$ **(c)** (`twobody_cont_phase.py`, `twobody_cont_grid.py`).

Bad-Boussinesq / Toda Fermi-edge dictionary. From `phase2/toda_route.tex` eqs. (13)–(24) and `phase2/onebody_dispersion_test.tex` eqs. (30)–(36), the hyperbolic Toda chain with potential $V(r) = e^{-r} + r - 1$ at finite background strain $a > 0$ ($b = e^{-a/2}$, sound speed $c_s = a\sqrt{b}$) maps in its right-moving plus left-moving continuum limit to the bad Boussinesq equation (1) via **(c)** (`twobody_cont_grid.py`):

$$\begin{aligned} &\kappa_j = \sqrt{3}\,\beta_j, \qquad h = 3b\,\hbar_B, \qquad \hbar_B = \frac{h}{3b}, \\ &y_j = \frac{b\kappa_j^2}{h} = \frac{\beta_j^2}{\hbar_B}, \qquad \beta_j = \sqrt{\hbar_B\, y_j}, \qquad r := \frac{\beta_2}{\beta_1} = \sqrt{\frac{y_2}{y_1}}, \\ &P_B = \frac{a\,\overline{P}}{6\sqrt{3}\,b}, \qquad E_B = \frac{\overline{E}}{6\sqrt{3}\,b^2}, \qquad S_T = 3b\,S_B \implies \frac{\delta_T}{h} = \frac{\delta_B}{\hbar_B}. \end{aligned} \tag{3}$$

Because $S_T/h = S_B/\hbar_B$ holds identically, dimensionless semiclassical scattering phases $\delta(\beta_1, \beta_2) = \hbar_B^{-1}\delta_{0,B} + \delta_{1,B} + O(\hbar_B)$ require *no overall rescaling factor* between the bad Boussinesq continuum ($O(\hbar_B^0)$) and the Toda lattice ($O(h^0)$) **(c)** (`twobody_cont_grid.py`). In the Fermi-edge scaling regime where $y_j = \beta_j^2/\hbar_B$ is held fixed as $\hbar_B \to 0$, expanding at small amplitude $\beta_1 = \varepsilon > \beta_2 = r\varepsilon$ ($\varepsilon = \sqrt{\hbar_B y_1}$) organises the continuum phase into a power series in $\varepsilon^2 = y_1\hbar_B$.

2 Bad-Equation Two-Soliton Phase in the Lattice-Matched Scheme (Task 1)

2.1 Exact Factored Hirota Ratio and Classical Phase on the Bad Branch

Proposition 2.1 (**(c)**, `twobody_cont_phase.py`, `twobody_cont_grid.py`)**.** *On the bad Boussinesq branch ($k_j = \sinh\beta_j$, $v_j = \epsilon_j\cosh\beta_j$, $w_j = k_j v_j$, $\ell_j = \epsilon_j\cosh 3\beta_j$, $M_j = 2\sinh\beta_j$), the single-soliton dispersion relation $P_-(k_j, w_j) = w_j^2 - k_j^2 - k_j^4 = 0$ factors a common $2k_1k_2$ divisor out of the Hirota interaction coefficient $A_{12,b} = -P_-(k_1 - k_2, w_1 - w_2)/P_-(k_1 + k_2, w_1 + w_2)$, reducing it to the exact degree-2 form*

$$A_{12,b}(\beta_1, \epsilon_1; \beta_2, \epsilon_2) = \frac{\epsilon_1\epsilon_2\cosh\beta_1\cosh\beta_2 - 1 - 2(\sinh^2\beta_1 + \sinh^2\beta_2) + 3\sinh\beta_1\sinh\beta_2}{\epsilon_1\epsilon_2\cosh\beta_1\cosh\beta_2 - 1 - 2(\sinh^2\beta_1 + \sinh^2\beta_2) - 3\sinh\beta_1\sinh\beta_2}. \tag{4}$$

*Unlike the good branch (which has resonant poles at $\alpha_1 \mp \alpha_2 = \pi/3$), $A_{12,b}(\beta_1, \epsilon_1; \beta_2, \epsilon_2) > 0$ for all $\beta_1 \neq \beta_2 > 0$ in both the overtaking ($\epsilon_1 = \epsilon_2$) and head-on ($\epsilon_1 = -\epsilon_2$) channels **(c)**. For $v_1 > v_2$ (which covers overtaking with $\beta_1 > \beta_2$, $\epsilon_1 = \epsilon_2 = +1$ and head-on with $\epsilon_1 = +1, \epsilon_2 = -1$), the classical action phase shift is*

$$\delta_{0,b}(\beta_1, \epsilon_1; \beta_2, \epsilon_2) = \frac{2}{9}(\ell_1 - \ell_2)\ln A_{12,b} + \frac{2}{3}M_1M_2(v_1 - v_2), \tag{5}$$

and in all four sign sectors $(\epsilon_1, \epsilon_2) \in \{\pm 1\}^2$ its canonical momentum derivatives satisfy identically

$$-\frac{\partial\delta_{0,b}}{\partial P_{0,1}} = \Delta X_1' = \Delta x_{12} + \gamma_1 M_2, \qquad -\frac{\partial\delta_{0,b}}{\partial P_{0,2}} = \Delta X_2' = \Delta x_{21} - \gamma_2 M_1, \tag{6}$$

where $\Delta x_{12} = -\frac{\ln A_{12,b}}{\sinh\beta_1}$, $\Delta x_{21} = +\frac{\ln A_{12,b}}{\sinh\beta_2}$, and $\gamma_j = \frac{\mathrm{d}\Pi_j}{\mathrm{d}P_{0,j}} = -\frac{3\cosh 2\beta_j}{\sinh\beta_j\sinh 3\beta_j}$ (independent of ϵ_j).

In the small-amplitude corner $\beta_1 = \varepsilon > \beta_2 = r\varepsilon$ ($0 < r < 1$), Proposition 2.1 expands as **(c)** (`twobody_cont_phase.py`):

- **Overtaking channel** ($\epsilon_1 = \epsilon_2 = +1$, $v_1 > v_2 > 1$):

$$\frac{A_{\mathrm{ov}}(\varepsilon, r\varepsilon)}{A_0(r)} = 1 + r\varepsilon^2 + \frac{1}{2}r^2\varepsilon^4 + O(\varepsilon^6), \qquad A_0(r) := \left(\frac{1-r}{1+r}\right)^2, \tag{7}$$

$$\delta_{0,b}^{\mathrm{ov}}(\varepsilon, r\varepsilon) = (1 - r^2)\ln A_0(r)\,\varepsilon^2 + \left[\frac{3}{4}(1 - r^4)\ln A_0(r) + \frac{7}{3}r(1 - r^2)\right]\varepsilon^4 + O(\varepsilon^6). \tag{8}$$

- **Head-on channel** ($\epsilon_1 = +1, \epsilon_2 = -1$, $v_1 > 0 > v_2$):

$$A_{\rm ho}(\varepsilon, r\varepsilon) = 1 - 3r\varepsilon^2 + \left[\frac{9}{2}r^2 + \frac{13}{4}r(1+r^2)\right]\varepsilon^4 + O(\varepsilon^6), \qquad \ln A_{\rm ho} = -3r\varepsilon^2 + \frac{13}{4}r(1+r^2)\varepsilon^4 + O(\varepsilon^6), \tag{9}$$

$$\delta^{\rm ho}_{0,b}(\varepsilon, r\varepsilon) = 4r\varepsilon^2 + \frac{2}{3}r(1+r^2)\varepsilon^4 + O(\varepsilon^6). \tag{10}$$

Notice that in the head-on channel the Hirota trajectory shift is $\Delta x^{\rm ho}_{12} = -\frac{\ln A_{\rm ho}}{\sinh\beta_1} = +3r\varepsilon + O(\varepsilon^3)$, which is $O(\varepsilon^2)$ smaller than the Darboux tail shift $\gamma_1 M_2 = -\frac{2r}{\varepsilon} + O(\varepsilon)$, so the leading classical head-on phase $\delta^{\rm ho}_{0,b} = 4\beta_1\beta_2 + O(\varepsilon^4)$ arises from adding $-\frac{4}{3}\beta_1\beta_2$ (from $\frac{2}{9}(\ell_1 - \ell_2)\ln A_{\rm ho}$) and $+\frac{16}{3}\beta_1\beta_2$ (from $\frac{2}{3}M_1M_2(v_1 - v_2)$) **(c)** (`twobody_cont_phase.py`).

2.2 Phonon Determinant Phase Integrals I_{12} and J_{12}

Following `phase2/oneloop_phase.tex`, `phase2/errata.tex`, and `phase2/audit_determinant_foundations.tex`, let $I_{12} = \frac{1}{4\pi}\int_{\mathbb{R}\setminus\{0\}} \theta_1(q)\theta_2'(q)\,{\rm d}q$ be the raw stationary-phase phonon integral and $J_{12} = \frac{1}{2}(I_{12} - I_{21})$ its antisymmetrised partner. Because the $q = 0^{\pm}$ Levinson endpoint constants on the light good branch ($\alpha_j \in (0, \pi/3)$) are $\theta_j(0^{\pm}; \alpha_j, \epsilon_j) = \pi(\pm 1 + \epsilon_j)$ **(c)** (`phase2/audit_determinant_foundations.tex`), integration by parts yields **(c)**:

$$I_{12} + I_{21} = -\frac{\pi}{2}(\epsilon_1 + \epsilon_2), \qquad I_{12} = J_{12} - \frac{\pi}{4}(\epsilon_1 + \epsilon_2). \tag{11}$$

On the bad branch ($\alpha_j = -i\beta_j$), the three Lax roots are complex conjugates of one real root:

$$\kappa_{j,1} = \frac{1}{2}\sinh\beta_j + i\frac{\sqrt{3}}{2}\cosh\beta_j, \qquad \kappa_{j,2} = \overline{\kappa_{j,1}}, \qquad \kappa_{j,3} = -\sinh\beta_j, \tag{12}$$

with weights $(c_1, c_2, c_3) = (+1, +1, -2)$. Although the bad-Boussinesq vacuum is linearly unstable on the long-wavelength band $|q| \in (0, 1)$ (and $|q| \in (0, \sqrt{1 + \sinh^2\beta_j})$ around one soliton), the two-body scattering phase integral continued via $\alpha_j = -i\beta_j$ (or integrated directly along real $q \in \mathbb{R}\setminus\{0\}$ using the branch-continuous odd/even decomposition of $\theta_j(q)$ with $t = \sqrt{1+q^2}$) is real and UV-convergent for all $\beta_1 \neq \beta_2 > 0$ **(c)** (`twobody_cont_phase.py`).

Proposition 2.2 (**(c)**, `twobody_cont_phase.py`)**.** *In the small-amplitude corner $\beta_1 = \varepsilon > \beta_2 = r\varepsilon$ ($0 < r < 1$) on the bad branch:*

1. ***Overtaking channel*** *($\epsilon_1 = \epsilon_2 = +1$): The $r = 3, s = 3$ real-root sector ($\kappa_{j,3} = -\sinh\beta_j$) reproduces the scale-invariant KdV integral $I_{\rm KdV}(r)$ at $O(1)$, while the complex conjugate root pairs $\kappa_{j,1}, \kappa_{j,2}$ contribute starting at $O(\varepsilon^2)$:*

$$I^{\rm ov}_{12,b}(\varepsilon, r\varepsilon) = I_{\rm KdV}(r) + c^{\rm ov}_{2,\rm det}(r)\,\varepsilon^2 + O(\varepsilon^4), \qquad J^{\rm ov}_{12,b}(\varepsilon, r\varepsilon) = \frac{\pi}{2} + I^{\rm ov}_{12,b}(\varepsilon, r\varepsilon), \tag{13}$$

$$I_{\rm KdV}(r) = -\frac{4}{\pi}\chi_2(r) - \frac{2}{\pi}\ln r \ln\frac{1-r}{1+r} = \frac{4}{\pi}\int_0^r \frac{\ln x}{1-x^2}\,{\rm d}x, \tag{14}$$

$$c^{\rm ov}_{2,\rm det}(r) = -\frac{1}{3\pi}\left[2(1-r^2)\ln\frac{1-r}{1+r} + 9r\right], \tag{15}$$

where $\chi_2(r) = \frac{1}{2}[{\rm Li}_2(r) - {\rm Li}_2(-r)]$. Formula (15) is verified to 60 digits against direct 2D quadrature of the q-integral and to $O(\varepsilon^4)$ against the exact 9-term dilogarithm sum ***(c)*** *(`twobody_cont_phase.py`).*

2. ***Head-on channel*** *($\epsilon_1 = +1, \epsilon_2 = -1$): Both $I^{\rm ho}_{12,b}$ and $J^{\rm ho}_{12,b}$ vanish at $O(1)$ and start at $O(\varepsilon^2)$:*

$$I^{\rm ho}_{12,b}(\varepsilon, r\varepsilon) = J^{\rm ho}_{12,b}(\varepsilon, r\varepsilon) = c^{\rm ho}_{2,\rm det}(r)\,\varepsilon^2 + O(\varepsilon^4), \qquad c^{\rm ho}_{2,\rm det}(r) = -\frac{8}{\pi}r. \tag{16}$$

3. ***Invariance under $\Delta\mathcal{H}$:*** *Because the lattice-matching counterterm $\Delta\mathcal{H} = \hbar_B \lambda u^2$ ($\lambda = -\frac{3}{2\pi}$) is explicitly $O(\hbar_B)$, it enters the one-loop effective action directly at $O(\hbar_B)$ evaluated on the classical saddle $u_{\rm cl}$ and does not modify the $O(\hbar_B^0)$ quadratic fluctuation operator $\mathcal{D}_v$ or the phonon phase shift $\theta_j(q)$. Thus $I_{12,b}$ and $J_{12,b}$ are unchanged at $O(\hbar_B^0)$* ***(c).***

2.3 The Three Continuum Implementations of the Lattice-Matched Scheme at Two-Body

In `phase2/onebody_scheme.tex` and `phase2/audit_onebody_scheme.tex`, matching the continuum vacuum sound-speed elasticity to the Toda chain fixes the local counterterm

$$\Delta\mathcal{H} = \hbar_B \lambda u^2, \qquad \lambda = s_{1,T} - s_{1,b,\rm ac} = \frac{3}{4\pi} - \frac{9}{4\pi} = -\frac{3}{2\pi}. \tag{17}$$

On a stationary one-soliton background u_0, $\int_{-\infty}^{\infty} u_0^2 \,\mathrm{d}x = \frac{2}{3}\sinh^3\beta$, so $\Delta\mathcal{H}$ adds $\Delta_{\rm ct}F(\beta) = \frac{2}{3}\lambda\sinh^3\beta = -\frac{1}{\pi}\sinh^3\beta$ to the one-loop collective energy $F(\beta) = -L_1(\beta) = \Delta E|_P(\beta)/\hbar_B$, shifting **(c)** (`twobody_cont_phase.py`):

$$F_{\rm ac}(\beta) = \frac{\sqrt{3}}{\pi}\left(\sinh^2\beta + \frac{3}{4}\right)\arctan\frac{2\sinh\beta}{\sqrt{3}} - \frac{3}{2\pi}\sinh\beta = \frac{4}{3\pi}\beta^3 + \frac{14}{45\pi}\beta^5 + \frac{97}{1890\pi}\beta^7 + O(\beta^9), \tag{18}$$

$$F_{\rm matched}(\beta) = F_{\rm ac}(\beta) - \frac{1}{\pi}\sinh^3\beta = \frac{1}{3\pi}\beta^3 - \frac{17}{90\pi}\beta^5 - \frac{431}{7560\pi}\beta^7 + O(\beta^9), \tag{19}$$

$$C_{\rm matched} = \lim_{\beta\to 0}\frac{F_{\rm matched}(\beta)}{|P_0(\beta)|} = \frac{1}{2\pi} = C_T \qquad \left(\text{vs. } C_{\rm ac} = \frac{2}{\pi} = 4C_T\right). \tag{20}$$

As noted in `phase2/audit_onebody_scheme.tex` and `phase2/audit_smx.tex`, at the one-body level the fixed-momentum energy shift $\Delta E|_P = \hbar_B(e - vp) = -\hbar_B L_1$ is invariant under any redefinition $(e, p) \mapsto (e + v\delta p,\, p + \delta p)$ because $v\delta p$ cancels out of $e - vp$. At the **two-body** level, however, converting the fixed-parameter saddle phase into the fixed-physical-momentum scattering phase produces the Faddeev–Korepin frame shift

$$\delta_{\rm renorm}(1,2) = -\frac{\partial\delta_0}{\partial P_{0,1}}p_1 - \frac{\partial\delta_0}{\partial P_{0,2}}p_2 = \Delta X_1' p_1 + \Delta X_2' p_2 = H_{12}[p] + D_{12}[p], \tag{21}$$

where $P_{\rm phys,j} = P_{0,j} + \hbar_B p_j$, $H_{12}[p] = \Delta x_{12}p_1 + \Delta x_{21}p_2$ is the Hirota part, and $D_{12}[p] = \gamma_1 M_2 p_1 - \gamma_2 M_1 p_2$ is the Darboux tail part. Equation (21) depends on $p_j(\beta_j, \epsilon_j)$ alone! We therefore evaluate all three candidate implementations of the lattice-matched scheme in `twobody_cont_phase.py`:

Proposition 2.3 (**(c)**, `twobody_cont_phase.py`)**.** *In the lattice-matched scheme ($\lambda = -\frac{3}{2\pi}$), the three continuum fixed-momentum prescriptions give the following analytical expansions in the small-amplitude corner $\beta_1 = \varepsilon > \beta_2 = r\varepsilon$ ($0 < r < 1$):*

1. ***Prescription C-matched (Canonical collective momentum $p_{\rm can,matched} = \frac{\mathrm{d}L_{1,\rm matched}}{\mathrm{d}v}$):*** *Since $L_{1,\rm matched}(v) = -F_{\rm matched}(\beta(v))$ is the one-loop effective collective Lagrangian, Noether's theorem on the collective coordinate $X(t)$ defines the canonical momentum shift $p_{\rm can,matched}(\beta,\epsilon) = \frac{\epsilon}{\sinh\beta}\frac{\mathrm{d}L_{1,\rm matched}}{\mathrm{d}\beta}$:*

$$\begin{aligned} p_{\rm can,matched}(\beta,\epsilon) &= -\frac{2\epsilon\sqrt{3}}{\pi}\cosh\beta\arctan\frac{2\sinh\beta}{\sqrt{3}} + \frac{3\epsilon}{\pi}\sinh\beta\cosh\beta \\ &= -\frac{\epsilon}{\pi}\beta + \frac{10\epsilon}{9\pi}\beta^3 + \frac{2\epsilon}{9\pi}\beta^5 + O(\beta^7), \end{aligned} \tag{22}$$

$$p_{\rm can,ac}(\beta,\epsilon) = -\frac{2\epsilon\sqrt{3}}{\pi}\cosh\beta\arctan\frac{2\sinh\beta}{\sqrt{3}} = -\frac{4\epsilon}{\pi}\beta - \frac{8\epsilon}{9\pi}\beta^3 - \frac{4\epsilon}{15\pi}\beta^5 + O(\beta^7). \tag{23}$$

More generally, for any smooth $F(\beta) = \frac{2}{3}C\beta^3 + O(\beta^5)$, the linear slope of $p_{\rm can}$ is universally $c_1(p_{\rm can}) = -2C$, and the dressed momentum Jacobian satisfies

$$\frac{\mathrm{d}(P_0 + \hbar_B p_{\rm can,matched})}{\mathrm{d}P_0} = 1 - \frac{C_{\rm matched}}{y} + \frac{5\beta^2}{2\pi y} + O(\beta^4) = 1 - \frac{1}{2\pi y} + O(\hbar_B), \tag{24}$$

which matches the large-y Toda Bethe Jacobian $u'(y) = 1 - \frac{1}{2\pi y} + O(y^{-3/2})$ in ***phase2/frame_dictionary.t*** ***(c)*** *(**twobody_cont_det_antisym.py**). Because $c_{\ln} = 0$ in (22),* ***all*** *$\ln\varepsilon$* ***infrared divergences vanish****, H_{12} vanishes at $O(1)$, and D_{12} is finite and scale-invariant in both channels:*

$$H_{\rm ov}[p_{\rm can,matched}] = -\frac{23}{18\pi}(1-r^2)\ln A_0(r)\,\varepsilon^2 + O(\varepsilon^4) \xrightarrow{\varepsilon\to 0} 0, \tag{25}$$

$$D_{\rm ov}[p_{\rm can,matched}] = -\frac{2}{\pi}\left(r^{-1} - r\right) - \frac{1}{3\pi}\left(r^{-1} - r^3\right)\varepsilon^2 + O(\varepsilon^4) \qquad \left(vs.\ -\frac{8}{\pi}(r^{-1}-r)\ for\ p_{\rm can,ac}\right), \tag{26}$$

$$H_{\rm ho}[p_{\rm can,matched}] = -\frac{6}{\pi}r\varepsilon^2 + O(\varepsilon^4) \xrightarrow{\varepsilon\to 0} 0, \tag{27}$$

$$D_{\rm ho}[p_{\rm can,matched}] = +\frac{2}{\pi}\left(r^{-1} + r\right) + \frac{1}{\pi}\left[\frac{1}{3}\left(r^{-1} + r^3\right) - \frac{28}{9}r\right]\varepsilon^2 + O(\varepsilon^4) \qquad \left(vs.\ +\frac{8}{\pi}(r^{-1}+r)\ for\ p_{\rm can,ac}\right) \tag{28}$$

2. ***Prescription M (Fixed-profile mode-sum momentum*** *$p_{\rm mode}$****):*** *Because $\Delta\mathcal{L} = -\Delta\mathcal{H} = -\hbar_B\lambda\phi_x^2$ contains no time derivatives ϕ_t, the field momentum density $\mathcal{P} = -\pi\phi_x$ is independent of λ, so at fixed classical profile $\Delta_{\rm ct}p_{\rm mode} = 0$ (**phase2/audit_onebody_scheme.tex**). Thus $p_{\rm mode}$ retains its bare expansion*

$$p_{\rm mode}(\beta,\epsilon) = \epsilon\left(c_{\ln}\beta\ln\beta + c_{\rm lin}\beta\right) + O(\beta^3\ln\beta), \qquad c_{\ln} = \frac{1}{\pi}, \quad c_{\rm lin} = \frac{-\ln 2 - 1 + \frac{1}{4}\ln 3}{\pi}, \tag{29}$$

which yields a non-zero $O(1)$ Hirota shift in the overtaking channel that cancels the log-product term of $I_{\rm KdV}(r)$, alongside a scale-dependent $\ln\varepsilon$ infrared divergence in the Darboux shifts:

$$H_{\rm ov}[p_{\rm mode}] = +\frac{2}{\pi}\ln r\ln\frac{1-r}{1+r} + O(\varepsilon^2\ln\varepsilon), \tag{30}$$

$$D_{\rm ov}[p_{\rm mode}] = 2\left(r^{-1} - r\right)\left(\frac{1}{\pi}\ln\varepsilon + c_{\rm lin}\right) + \frac{2}{\pi r}\ln r + O(\varepsilon^2\ln\varepsilon), \tag{31}$$

$$H_{\rm ho}[p_{\rm mode}] = O(\varepsilon^2\ln\varepsilon) \to 0, \qquad D_{\rm ho}[p_{\rm mode}] = -2\left(r^{-1} + r\right)\left(\frac{1}{\pi}\ln\varepsilon + c_{\rm lin}\right) - \frac{2}{\pi r}\ln r + O(\varepsilon^2\ln\varepsilon). \tag{32}$$

3. ***Prescription S (Direct on-shell sound-speed rescaling*** *$c_s = \sqrt{1 + 2\hbar_B\lambda}$****):*** *Absorbing $\Delta\mathcal{H} = \hbar_B\lambda u^2$ into the modified equation of motion $u_{tt} = c_s^2 u_{xx} + u_{xxxx} + 3(u^2)_{xx}$ with $c_s^2 = 1 + 2\hbar_B\lambda$ rescales the exact N-soliton solutions via $x = c_s^{-1}\tilde{x}$, $t = c_s^{-2}\tilde{t}$, $\phi = c_s\tilde{\phi}$, $u = c_s^2\tilde{u}$, under which*

$$S_{c_s} = c_s^3\tilde{S}, \qquad P_{c_s} = c_s^4\tilde{P}, \qquad E_{c_s} = c_s^5\tilde{E}, \qquad v_{c_s} = c_s\tilde{v}. \tag{33}$$

Differentiating with respect to c_s^2 at $c_s = 1$ ($\frac{\mathrm{d}c_s}{\mathrm{d}(\hbar_B\lambda)} = 1$) at fixed physical momentum $P_{c_s} = P_0$ yields the exact one-soliton Virial identity $5E_0(\beta) - 4v(\beta)P_0(\beta) = \frac{2}{3}\sinh^3\beta = \int u_0^2\,\mathrm{d}x$ and the two-soliton fixed-momentum counterterm phase shift

$$\Delta_{\rm ct}^{\rm sound}\delta(1,2) = \lambda\left(3\delta_{0,b} + 4P_{0,1}\Delta X_1' + 4P_{0,2}\Delta X_2'\right). \tag{34}$$

Because the leading small-amplitude classical phase $\delta_{0,b}^{(0)} \propto |P_{0,1}|^{1/3}|P_{0,2}|^{1/3}\hat{\delta}_0((P_{0,2}/P_{0,1})^{1/3})$ *is homogeneous of degree* $\frac{2}{3} < \frac{3}{4}$ *in* $(P_{0,1}, P_{0,2})$, $\Delta_{\rm ct}^{\rm sound}\delta(1,2)$ *has* no $O(1)$ piece *and starts at* $O(\varepsilon^2)$ *in both channels:*

$$\Delta_{\rm ct}^{\rm sound}\delta^{\rm ov}(\varepsilon, r\varepsilon) = -\frac{1}{2\pi}(1-r^2)\ln A_0(r)\,\varepsilon^2 + O(\varepsilon^4) \xrightarrow{\varepsilon\to 0} 0, \tag{35}$$

$$\Delta_{\rm ct}^{\rm sound}\delta^{\rm ho}(\varepsilon, r\varepsilon) = -\frac{2}{\pi}r\varepsilon^2 - \frac{3}{\pi}r(1+r^2)\varepsilon^4 + O(\varepsilon^6) \xrightarrow{\varepsilon\to 0} 0. \tag{36}$$

3 Open Problem #2: Boundary-Compatible Determinant and Undetermined Terms (Task 2)

3.1 Exact Finite-R Kinetic Norm and Non-Commuting Projector Tails on the Bad Branch

In `phase2/det_projected_oneloop.tex` and `phase2/audit_det_projected.tex`, the instantaneous collective phase space was completed to a four-dimensional symplectic space (X, P, C, Π) with finite symplectic pairing $\Omega(\xi_X, \xi_P) = \frac{\mathrm{d}P_0}{\mathrm{d}v}$ and $\Omega(\xi_C, \xi_\Pi) = 1$, but the configuration-space kinetic norm of time-modulated shift and width variations was shown to diverge linearly with the box size R. We establish the exact bad-branch formula including the finite $O(1)$ remainder:

Proposition 3.1 (**(c)**, `twobody_cont_det_antisym.py`)**.** *For the bad-Boussinesq potential profile* $\Phi(\xi; v, C) = C - \sinh\beta\tanh(\frac{1}{2}\xi\sinh\beta)$ *with* $k = \sinh\beta$ *and* $v = \epsilon\cosh\beta$*:*

1. *The collective symplectic pairing is* $\Omega(\xi_X, \xi_P) = \frac{\mathrm{d}P_0}{\mathrm{d}v} = \frac{2}{3}\sinh 3\beta = \frac{2}{3}k(3+4k^2) > 0$ *for all* $\beta > 0$ ***(c)****.*

2. *The velocity tangent is* $\partial_v\Phi = -\epsilon\coth\beta\left(\tanh z + z\,\mathrm{sech}^2 z\right)$ *where* $z = \frac{1}{2}k\xi$*, with non-vanishing spatial limits* $\partial_v\Phi(\pm\infty) = \mp\epsilon\coth\beta = \mp\frac{v}{k}$*. On* $[-R, R]$*, using* $\int_{-\infty}^{\infty}[(\tanh z + z\,\mathrm{sech}^2 z)^2 - 1]\,\mathrm{d}z = 2\int_0^\infty z^2\,\mathrm{sech}^4 z\,\mathrm{d}z = +\frac{\pi^2-6}{9}$*, the configuration-space kinetic integral of a time-modulated profile* $\dot\Phi = \dot C + \dot v\,\partial_v\Phi$ *is*

$$\frac{1}{2}\int_{-R}^{R}\left(\dot C + \dot v\,\partial_v\Phi\right)^2\mathrm{d}\xi = R\left(\dot C^2 + \frac{v^2}{k^2}\dot v^2\right) + \frac{v^2(\pi^2-6)}{9k^3}\dot v^2 + O\Big(R^2 e^{-kR}\Big). \tag{37}$$

3. *For any localized smooth phase-space fluctuation* $\eta = (\delta\phi, \delta\pi)$*, the symplectic projector* $\mathcal{P} = I - \Pi_{\rm coll}$ *onto the orthogonal complement of the collective subspace produces non-vanishing spatial tails* $(\mathcal{P}\eta)_\phi(\pm\infty) = b_r \pm \frac{v}{k}b_X$*, and the non-zero collective Poisson brackets* $\{X, P\} = 1$*,* $\{C, \Pi\} = 1$ *imply* $[\partial_t - v\partial_\xi, \mathcal{P}] \neq 0$ ***(c)****.*

3.2 Cutoff-Endpoint and Subtraction Sensitivity of $L_1(\beta) = -F(\beta)$

As proved in `phase2/det_projected_oneloop.tex` eq. (30) and audited in `phase2/audit_det_projected.tex`, in a finite-cutoff phonon trace with upper endpoint $B_\Lambda = \Lambda + b_1 k + b_2 k^2 + b_3 k^3 + O(\Lambda^{-1})$ and finite subtraction C_Λ, the one-loop collective energy $F(\beta) = -L_1(\beta)$ acquires local polynomial terms in $k = \sinh\beta$:

$$F(\beta; c_{\rm tad}, \lambda, c_5) = F_{\rm ac}(\beta) + 2c_{\rm tad}\sinh\beta + \frac{2}{3}\lambda\sinh^3\beta + c_5\sinh^5\beta + O(\sinh^7\beta), \tag{38}$$

corresponding to local counterterms $c_{\rm tad}\int(-u_0)\,\mathrm{d}x = 2c_{\rm tad}\sinh\beta$, $\lambda\int u_0^2\,\mathrm{d}x = \frac{2}{3}\lambda\sinh^3\beta$, and $c_5\sinh^5\beta$ (from u_x^2 or u^3). In `twobody_cont_det_antisym.py` we verify **(c)**:

$$\frac{F(\beta; c_{\rm tad}, \lambda, c_5)}{|P_0(\beta)|} = \frac{3c_{\rm tad}}{\beta^2} + \left(\frac{2}{\pi} + \lambda\right) + \left(\frac{3}{2}c_5 - \frac{47}{60\pi}\right)\beta^2 + O(\beta^4) \qquad \left(\text{at } \lambda = -\frac{3}{2\pi}\right), \tag{39}$$

$$p_{\rm can}(\beta, \epsilon; c_{\rm tad}, \lambda, c_5) = -\frac{2\epsilon c_{\rm tad}}{\beta} + \epsilon\left(-\frac{4}{\pi} - 2\lambda\right)\beta + \epsilon\left(\frac{10}{9\pi} - 5c_5\right)\beta^3 + O(\beta^5). \tag{40}$$

From (39)–(40) we establish three structural facts **(c)** (`twobody_cont_det_antisym.py`):

1. A non-zero tadpole/endpoint-linear coefficient $c_{\rm tad} \neq 0$ produces a $\frac{3c_{\rm tad}}{\beta^2}$ singularity in $F/|P_0|$ and a $-\frac{2\epsilon c_{\rm tad}}{\beta}$ singularity in $p_{\rm can}$. It is set to zero ($c_{\rm tad} = 0$) by the equal-mode-count / zero-tadpole vacuum condition $\langle u\rangle_{\rm vac} = 0$; moreover, in the two-soliton determinant ratio $\det\mathcal{D}(u_{12})/[\det\mathcal{D}(u_1)\det\mathcal{D}(u_2)]$ any local tadpole term $\propto \int u\,{\rm d}x = -M$ cancels identically because $M_{12} = M_1 + M_2$.

2. The elastic u^2 counterterm coefficient λ is uniquely fixed to $\lambda = -\frac{3}{2\pi}$ by matching the vacuum sound-speed shift to the Toda lattice (`phase2/onebody_scheme.tex`).

3. Once $c_{\rm tad} = 0$ and $\lambda = -\frac{3}{2\pi}$ are fixed, **any local analytic higher-derivative counterterm** $c_5\sinh^5\beta + O(\sinh^7\beta)$ starts at $O(\beta^2)$ in $F/|P_0|$, starts at $O(\beta^3)$ in $p_{\rm can}$, and contributes only at $O(\varepsilon^2) \to 0$ to H_{12} and D_{12} in the small-amplitude corner.

4. By contrast, if one defines the bad-branch one-body trace by restricting the q-integral to the real stable band $|q| > \sqrt{1+\sinh^2\beta}$, the resulting energy $F_{\rm band}(\beta) = F_{\rm ac}(\beta) - \frac{\sqrt{3}}{2}(\sinh^2\beta + \frac{3}{4})$ has a non-zero $O(1)$ threshold constant $F_{\rm band}(0) = -\frac{3\sqrt{3}}{8}$ from cutting out the unstable window $|q| \in (0, \sqrt{1+\sinh^2\beta})$ **(c)** (`twobody_cont_det_antisym.py`, confirming `phase2/onebody_dispersion_test.tex`).

3.3 Precise Statement of What Remains Undetermined After Lattice-Matched Renormalisation

Combining Propositions 2.3 and 3.1 with the audit verdicts (`phase2/audit_det_projected.tex`, `phase2/audit_determinant_foundations.tex`, `phase2/audit_onebody_scheme.tex`, `phase2/audit_smx.te`
we state precisely which constants or terms remain undetermined on the continuum side after fixing $\Delta\mathcal{H} = -\frac{3\hbar_B}{2\pi}u^2$:

1. **One-body non-commuting projector / boundary Schur complement $\Delta L_{1,\rm Schur}(\beta)$ (d):** Because $[\partial_t - v\partial_\xi, \mathcal{P}] \neq 0$ and $(\mathcal{P}\eta)_\phi(\pm\infty) \neq 0$ (Proposition 3.1), the time-dependent Gaussian integral over projected fluctuations on a finite-action history with compensating endpoint tails differs from the adiabatic phonon trace by a boundary Schur-complement determinant. While any *local analytic* counterterm starting at $O(\sinh^5\beta)$ has zero effect on $C_{\rm matched} = \frac{1}{2\pi}$ and zero $O(1)$ effect on H_{12}, D_{12} in the small-amplitude corner **(c)**, a non-local or threshold-singular boundary contribution is not ruled out by the instantaneous phase-space projector alone **(d)**.

2. **Two-body overall determinant normalisation constant $Z = e^{i\delta_{\rm const}}$ (I_{12} vs. J_{12}) (d):** Because the $q = 0^\pm$ Levinson endpoint values $\theta_j(0^\pm)$ are piecewise constant in each regular chamber, the Faddeev–Korepin gradient identities $\nabla_\beta I_{12} = \nabla_\beta J_{12}$ cannot fix the chamber-constant offset $J_{12} - I_{12} = \frac{\pi}{4}(\epsilon_1 + \epsilon_2)$ ($+\frac{\pi}{2}$ in the overtaking channel, 0 in the head-on channel) without an independent time-boundary / Maslov determinant normalisation (`phase2/audit_determinant_foundations.tex`).

3. **Two-body collective momentum shift prescription in $\delta_{\rm renorm} = \Delta X_1'\Delta P_1 + \Delta X_2'\Delta P_2$ (d):** As shown in Proposition 2.3, the one-body identity $e - vp = -L_1$ leaves open whether the one-soliton momentum shift entering the two-body frame conversion is the canonical collective momentum $p_{\rm can,matched}$ (giving $D_{\rm ov} = -\frac{2}{\pi}(r^{-1} - r)$, $D_{\rm ho} = +\frac{2}{\pi}(r^{-1} + r)$, $H_{12} = 0$), the mode-sum momentum $p_{\rm mode}$ (giving $H_{\rm ov} = +\frac{2}{\pi}\ln r \ln\frac{1-r}{1+r}$ and an infrared $\ln\varepsilon$ divergence in D_{12}), or whether the Darboux tail shift is cancelled by the two-soliton zero-mode Wronskian measure leaving only the sound-rescaled shift $\Delta_{\rm ct}^{\rm sound}\delta = O(\varepsilon^2) \to 0$.

4 Symmetric vs. Antisymmetric Separation and the Residual Unknown $\vartheta(1,2)$ (Task 3)

4.1 Exact Exchange Decomposition in the Symmetric-Cutoff Darboux Frame

Under full particle exchange $1 \leftrightarrow 2$, meaning $(\beta_1, \epsilon_1) \leftrightarrow (\beta_2, \epsilon_2)$ (which sends $r = \beta_2/\beta_1 \mapsto 1/r$ in the overtaking channel and swaps the right-mover and left-mover labels in the head-on channel), any two-body phase decomposes into $\delta_1(1,2) = \delta_1^{\mathrm{sym}}(1,2) + \delta_1^{\mathrm{anti}}(1,2)$ with $\delta_1^{\mathrm{sym}}(2,1) = +\delta_1^{\mathrm{sym}}(1,2)$ and $\delta_1^{\mathrm{anti}}(2,1) = -\delta_1^{\mathrm{anti}}(1,2)$.

Proposition 4.1 (**(c)**, `twobody_cont_phase.py`, `twobody_cont_det_antisym.py`)**.** *In the symmetric-cutoff Darboux frame:*

1. ***Phonon determinant integral:*** *For all $r > 0$ ($r \neq 1$), the KdV integral* (14) *satisfies the exact inversion identity*

$$I_{\mathrm{KdV}}(r) + I_{\mathrm{KdV}}(1/r) = -\pi, \qquad J_{\mathrm{KdV}}(r) = \frac{\pi}{2} + I_{\mathrm{KdV}}(r) = \frac{I_{\mathrm{KdV}}(r) - I_{\mathrm{KdV}}(1/r)}{2} = -J_{\mathrm{KdV}}(1/r). \tag{41}$$

   More generally, at any finite (β_1, β_2), the raw integral I_{12} has symmetric part $I_{12}^{\mathrm{sym}} = -\frac{\pi}{4}(\epsilon_1 + \epsilon_2)$ ($-\frac{\pi}{2}$ for overtaking $++$, 0 for head-on $+-$) and antisymmetric part $I_{12}^{\mathrm{anti}} = J_{12}$ ***(c)****.*

2. ***Frame conversion shifts:*** *Because $\Delta x_{21}(\beta_2, \epsilon_2; \beta_1, \epsilon_1) = -\Delta x_{12}(\beta_1, \epsilon_1; \beta_2, \epsilon_2)$ and the Darboux shift $D_{12}[p] = \gamma_1 M_2 p_1 - \gamma_2 M_1 p_2$ reverses sign when* (1) $\leftrightarrow$ (2)*, both $H_{12}[p]$ and $D_{12}[p]$ (for* every *one-body shift $p(\beta, \epsilon)$) as well as $\Delta_{\mathrm{ct}}^{\mathrm{sound}}\delta(1,2)$ in* (34) *are* ***strictly antisymmetric*** *under full particle exchange $(\beta_1, \epsilon_1) \leftrightarrow (\beta_2, \epsilon_2)$ at all orders in β_1, β_2* ***(c)****:*

$$H_{21}[p] = -H_{12}[p], \qquad D_{21}[p] = -D_{12}[p], \qquad \Delta_{\mathrm{ct}}^{\mathrm{sound}}\delta(2,1) = -\Delta_{\mathrm{ct}}^{\mathrm{sound}}\delta(1,2). \tag{42}$$

   In particular, in the overtaking small-amplitude limit, $D_{\mathrm{ov}}[p_{\mathrm{can,matched}}](r) = -\frac{2}{\pi}(r^{-1} - r)$ is manifestly odd under $r \mapsto 1/r$. In the head-on channel ($\epsilon_1 = +1, \epsilon_2 = -1$), swapping (1) $\leftrightarrow$ (2) *sends $(\beta_1, +1; \beta_2, -1) \mapsto (\beta_2, -1; \beta_1, +1)$; since $p(\beta, \epsilon) = \epsilon\, p(\beta, +1)$ is odd in ϵ, $D_{\mathrm{ho}}[p](\beta_1, +1; \beta_2, -1) = (\gamma_1 M_2 p(\beta_1, +1) + \gamma_2 M_1 p(\beta_2, +1))$ is symmetric in (β_1, β_2) at fixed signs $(+1, -1)$ and therefore* antisymmetric *when the signs ϵ_1, ϵ_2 are swapped along with the amplitudes* ***(c)*** *(`twobody_cont_phase.py`).*

4.2 Directional Frame Decomposition and Lattice-Matched Prediction for the Toda Mixed Rectangle $\mathcal{D}_0(Y)$

Why does a non-trivial *symmetric* piece appear in the Toda finite-sea frame analysis (`phase2/frame_dictionary` `phase2/audit_frame_dictionary.tex`)? Because on a finite open chain of length N ordered from left to right, the collective / Bethe coordinate transformation is *directional*: a soliton at position X_j shifts the phase of solitons to its right differently from solitons to its left ($\tilde{X}_1 = X_1$, $\tilde{X}_2 = X_2 - \gamma_2 M_1$ rather than the symmetrised $\pm\frac{1}{2}\gamma_j M_{3-j}$). Consequently, as derived in `phase2/frame_dictionary.tex` eqs. (144), (247)–(248), and (303)–(308), the directional frame conversion $C_{\mathrm{cl}}(1,2) = C_{\mathrm{cl}}^{\mathrm{sym}}(1,2) + C_{\mathrm{cl}}^{\mathrm{anti}}(1,2)$ acquires a symmetric cross term $S_h(y,z)$ built from the one-body momentum Jacobian $u'(y) = \frac{\mathrm{d}P_{\mathrm{phys}}}{\mathrm{d}P_{\mathrm{bare}}}$ and the sea-edge endpoint responses.

In `twobody_cont_det_antisym.py`, we combine the exact finite-sea endpoint identity of `phase2/frame_dictionary.tex` eq. (144),

$$2at\,\rho(q)\eta(q) = \frac{2a}{t}D(q)V(q) = \frac{2a}{2\pi h}\cdot\frac{\pi h}{a} = 1, \tag{43}$$

with the continuum one-loop Jacobian expansion $u'(y) = 1 - \frac{C}{y} + O(y^{-3/2})$, which gives $-\frac{2\sqrt{y}}{u'(y)} = -2\sqrt{y} - \frac{2C}{\sqrt{y}} + O(y^{-1})$. Applying the mixed second difference (rectangle) operator

$\mathcal{D}_0(Y) := S_0(2Y, Y/2) - S_0(Y, Y/2) - S_0(2Y, Y/4) + S_0(Y, Y/4)$ annihilates all additive one-body terms (including the sea-edge endpoint constants) and isolates the multiplicative cross term $8\left[\frac{\sqrt{y}}{u'(y)}\frac{\sqrt{z}}{u'(z)} - \sqrt{yz}\right]$:

Proposition 4.2 ((c), `twobody_cont_det_antisym.py`). *For any one-body dispersion with large-y Jacobian $u'(y) = 1 - \frac{C}{y} + O(y^{-3/2})$, the symmetric mixed rectangle $\mathcal{D}_0(Y)$ has the exact large-Y expansion*

$$\mathcal{D}_0(Y) = -2C\left(15\sqrt{2} - 20\right) + \frac{8C^2(\sqrt{2}-1)^2}{Y} + O(Y^{-3/2}). \tag{44}$$

*Evaluating (44) in the **lattice-matched scheme** ($C_{\text{matched}} = \frac{1}{2\pi}$) versus the bare analytically continued scheme ($C_{\text{ac}} = \frac{2}{\pi}$) gives*

$$\lim_{Y\to\infty} \mathcal{D}_0^{\text{matched}}(Y) = -\frac{15\sqrt{2}-20}{\pi} \approx -0.38617465, \qquad \lim_{Y\to\infty} \mathcal{D}_0^{\text{ac}}(Y) = -\frac{4(15\sqrt{2}-20)}{\pi} \approx -1.54469859. \tag{45}$$

*Comparing against the read-only Toda finite-sea data in `phase2/scripts/frame_analysis.json` (Table 1), the lattice-matched continuum prediction agrees with $\mathcal{D}_0^{\text{Toda}}(32) = -0.39575 \pm 0.0217$ to within 0.0096 (well inside the Toda edge spread 0.0217), whereas the bare continued value is off by a factor of 4 (residual 1.149) **(c)**.*

Table 1: Comparison of the continuum lattice-matched ($C_{\text{matched}} = \frac{1}{2\pi}$) and bare continued ($C_{\text{ac}} = \frac{2}{\pi}$) predictions for the symmetric mixed rectangle $\mathcal{D}_0(Y)$ against the Toda finite-sea data from `frame_analysis.json` **(c)** (`twobody_cont_det_antisym.py`).

Y	Toda $\mathcal{D}_0(Y)$	Toda edge spread	Factorised $u'(y)$	Cont. matched ($Y\to\infty$)	Cont. bare ac ($Y\to\infty$)
4.0	−0.504362	0.000365	−0.504365	−0.386175	−1.544699
8.0	−0.434983	0.001323	−0.435004	−0.386175	−1.544699
16.0	−0.408705	0.005203	−0.408882	−0.386175	−1.544699
32.0	−0.395752	0.021669	−0.397257	−0.386175	−1.544699

4.3 The Antisymmetric Residual Phase $\vartheta(1,2)$ and the Missing Ingredient

In `phase2/smx_candidates.tex` and `phase2/audit_smx.tex`, the full quantum two-soliton scattering phase at fixed physical momenta is written in terms of the classical action phase $\delta_0/\hbar_B$, the symmetric frame / Stirling pieces, and a single residual antisymmetric $O(\hbar_B^0)$ function $\vartheta(1,2) = -\vartheta(2,1)$ (or $\delta_1^{\text{anti}}(1,2)$). From Propositions 2.2, 2.3, and 4.1, the continuum antisymmetric $O(\hbar_B^0)$ phase in the small-amplitude corner ($r = \beta_2/\beta_1 = \sqrt{y_2/y_1} \in (0,1)$) is assembled from two well-defined building blocks:

$$\delta_1^{\text{anti}}(1,2) = J_{12}(1,2) + H_{12}[\Delta P] + D_{12}[\Delta P], \tag{46}$$

which takes the following explicit forms across the three continuum momentum prescriptions **(c)** (`twobody_cont_det_antisym.py`, `twobody_cont_grid.py`):

1. **Candidate C-matched (Lattice-matched canonical momentum $p_{\text{can,matched}}$, scale-invariant):**

$$\delta_{1,\text{ov}}^{\text{anti,[C-matched]}}(r) = J_{\text{KdV}}(r) + D_{\text{ov}}[p_{\text{can,matched}}](r) = \frac{\pi}{2} - \frac{4}{\pi}\chi_2(r) - \frac{2}{\pi}\ln r \ln\frac{1-r}{1+r} - \frac{2}{\pi}\left(r^{-1} - r\right), \tag{47}$$

$$\delta_{1,\text{ho}}^{\text{anti,[C-matched]}}(r) = D_{\text{ho}}[p_{\text{can,matched}}](r) = +\frac{2}{\pi}\left(r^{-1} + r\right). \tag{48}$$

(If the raw integral $I_{\mathrm{KdV}}(r) = J_{\mathrm{KdV}}(r) - \frac{\pi}{2}$ is used in place of $J_{\mathrm{KdV}}(r)$ on the $0 < r < 1$ chamber, subtract $\frac{\pi}{2}$ from (47).)

2. **Candidate S / Phonon-Only (Darboux tail shift cancelled by zero-mode Wronskian measure, or sound-rescaled $\Delta_{\mathrm{ct}}^{\mathrm{sound}}\delta = O(\varepsilon^2) \to 0$):**

$$\delta_{1,\mathrm{ov}}^{\mathrm{anti,[S]}}(r) = J_{\mathrm{KdV}}(r) = \frac{\pi}{2} - \frac{4}{\pi}\chi_2(r) - \frac{2}{\pi}\ln r \ln\frac{1-r}{1+r}, \qquad \delta_{1,\mathrm{ho}}^{\mathrm{anti,[S]}}(r) = 0. \tag{49}$$

3. **Candidate M (Mode-sum momentum p_{mode}, scale-dependent):**

$$\delta_{1,\mathrm{ov}}^{\mathrm{anti,[M]}}(r;\varepsilon) = \frac{\pi}{2} - \frac{4}{\pi}\chi_2(r) + 2\left(r^{-1} - r\right)\left(\frac{1}{\pi}\ln\varepsilon + c_{\mathrm{lin}}\right) + \frac{2}{\pi r}\ln r, \tag{50}$$

$$\delta_{1,\mathrm{ho}}^{\mathrm{anti,[M]}}(r;\varepsilon) = -2\left(r^{-1} + r\right)\left(\frac{1}{\pi}\ln\varepsilon + c_{\mathrm{lin}}\right) - \frac{2}{\pi r}\ln r. \tag{51}$$

Why $\vartheta(1,2)$ cannot be uniquely closed from the existing continuum ingredients alone (c)/(d). Selecting unconditionally among (47), (49), and (50) on the continuum side requires one missing continuum ingredient: *a time-dependent two-soliton collective-coordinate path-integral reduction in the potential field ϕ that simultaneously tracks (i) the compensating spatial endpoint tails of Proposition 3.1 during the collision and (ii) the ratio of the two-soliton zero-mode Wronskian determinant to the product of one-soliton zero-mode norms in the Darboux coordinates (X_1', X_2').* As established in `phase2/audit_determinant_foundations.tex` eq. (9), the asymptotic two-soliton collective symplectic block has non-vanishing tail cross-pairing $\Omega(\partial_{\alpha_1}, \partial_{\alpha_2}) \to -M_1'\Pi_2'$, which is precisely the classical generator of the Darboux shift D_{12}. Whether the quantum Jacobian of transforming from the naive collective basis to the asymptotic Darboux basis cancels D_{12} (leaving Candidate S) or retains $D_{12}[p_{\mathrm{can,matched}}]$ (giving Candidate C-matched) is not determined by one-soliton data or by the static projected operator **(d)**. Tabulating each well-defined piece separately in `twobody_cont_grid.json` allows the coordinator's comparison with the parallel Toda two-body calculation to test each candidate directly.

5 Tabulated Continuum Grids and Error Budget (Task 4)

All well-defined continuum two-soliton phase components have been computed at 55-digit working precision in `twobody_cont_grid.py` and saved to `phase2/scripts/twobody_cont_grid.json`. Table 2 displays representative values on the small-amplitude ratio grid $r \in (0,1)$, together with the exact r-derivatives at $r = 0.5$:

$$\begin{aligned}
\left.\frac{\mathrm{d}I_{\mathrm{KdV}}(r)}{\mathrm{d}r}\right|_{r=0.5} &= \left.\frac{4\ln r}{\pi(1-r^2)}\right|_{r=0.5} = -\frac{16\ln 2}{3\pi} \approx -1.17680872,\\
\left.\frac{\mathrm{d}D_{\mathrm{ov}}[p_{\mathrm{can,matched}}](r)}{\mathrm{d}r}\right|_{r=0.5} &= \left.\frac{2}{\pi}\left(r^{-2}+1\right)\right|_{r=0.5} = +\frac{10}{\pi} \approx +3.18309886,\\
\left.\frac{\mathrm{d}\left(I_{\mathrm{KdV}} + D_{\mathrm{ov}}[p_{\mathrm{can,matched}}]\right)}{\mathrm{d}r}\right|_{r=0.5} &= \frac{30 - 16\ln 2}{3\pi} \approx +2.00629014.
\end{aligned} \tag{52}$$

Notice that at $r = 0.5$, the bare continued slope $\frac{\mathrm{d}}{\mathrm{d}r}(I_{\mathrm{KdV}} + D_{\mathrm{ov}}[p_{\mathrm{can,ac}}])|_{r=0.5} \approx +11.555587$ was dominated by the unrenormalised Darboux term $+\frac{40}{\pi} \approx +12.7324$ (`phase2/matching.tex`); in the lattice-matched scheme ($C_{\mathrm{matched}} = \frac{1}{2\pi}$) this Darboux slope is reduced by a factor of 4 to $+\frac{10}{\pi} \approx +3.1831$ **(c)** (`twobody_cont_grid.py`).

Table 3 displays representative values from the non-perturbative finite-amplitude bad-branch grid (β_1, β_2) in `twobody_cont_grid.json`, illustrating the $O(\varepsilon^2)$ convergence of $\delta_{\mathrm{renorm}}[p_{\mathrm{can,matched}}] = H_{12} + D_{12}$ toward the small-amplitude limits $D_{\mathrm{ov}}^{\text{C-m}}(0.5) = -3/(2\pi) \approx -0.954930$ and $D_{\mathrm{ho}}^{\text{C-m}}(0.5) = +5/(2\pi) \approx +1.591549$ as $\beta_1 \to 0$ at fixed $r = \beta_2/\beta_1 = 0.5$.

Table 2: Small-amplitude continuum $O(\hbar_B^0)$ phase components as functions of $r = \beta_2/\beta_1 = \sqrt{y_2/y_1} \in (0,1)$ in the overtaking ($\epsilon_1 = \epsilon_2 = +1$) and head-on ($\epsilon_1 = +1, \epsilon_2 = -1$) channels, plus exact $\mathrm{d}/\mathrm{d}r$ derivatives at $r = 0.5$ **(c)** (`twobody_cont_grid.py`). Numerical evaluation uncertainty is $< 10^{-50}$; truncating the small-amplitude expansion at $O(1)$ has an explicit $O(\varepsilon^2 = y_1\hbar_B)$ error budget given by the subleading coefficients in Eqs. (13), (16), and (25)–(28).

r	$\hat{\delta}_{0,\mathrm{ov}}(r)$	$I_{\mathrm{KdV}}(r)$	$J_{\mathrm{KdV}}(r)$	$-\frac{4}{\pi}\chi_2(r)$	$D_{\mathrm{ov}}^{\text{C-m}}(r)$	$I_{\mathrm{KdV}} + D_{\mathrm{ov}}^{\text{C-m}}$	$J_{\mathrm{KdV}} + D_{\mathrm{ov}}^{\text{C-m}}$	$D_{\mathrm{ho}}^{\text{C-m}}(r)$
0.10	−0.397328	−0.421623	+1.149173	−0.127466	−6.302536	−6.724159	−5.153363	+6.429860
0.20	−0.778493	−0.671236	+0.899560	−0.255796	−3.055775	−3.727011	−2.156214	+3.310423
0.25	−0.957798	−0.771397	+0.799399	−0.320572	−2.387324	−3.158721	−1.587925	+2.705634
0.30	−1.125164	−0.859994	+0.710803	−0.386035	−1.931080	−2.791074	−1.220277	+2.313052
0.40	−1.419101	−1.013172	+0.557624	−0.518919	−1.336902	−2.350073	−0.779277	+1.846197
0.49	−1.629584	−1.129047	+0.441750	−0.642077	−0.987279	−2.116326	−0.545529	+1.611167
0.50	−1.647918	−1.140921	+0.429875	−0.656135	−0.954930	−2.095851	−0.525054	+1.591549
0.51	−1.665279	−1.152583	+0.418213	−0.670281	−0.923845	−2.076428	−0.505632	+1.573192
0.60	−1.775313	−1.250225	+0.320572	−0.799399	−0.679061	−1.929286	−0.358489	+1.443038
0.70	−1.769000	−1.344734	+0.226062	−0.951636	−0.463823	−1.808557	−0.237761	+1.355085
0.75	−1.701476	−1.388489	+0.182307	−1.032107	−0.371362	−1.759851	−0.189055	+1.326391
0.80	−1.582005	−1.430439	+0.140358	−1.116634	−0.286479	−1.716918	−0.146121	+1.305071
0.90	−1.118873	−1.510567	+0.060230	−1.304641	−0.134396	−1.644963	−0.074166	+1.280312
$\mathrm{d}/\mathrm{d}r\|_{0.5}$	−1.784839	−1.176809	−1.176809	−1.398799	+3.183099	+2.006290	+2.006290	−1.909859

Table 3: Exact finite-amplitude bad-branch classical phase $\delta_{0,b}$, Hirota shift $H_{12}[p_{\mathrm{can,matched}}]$, Darboux shift $D_{12}[p_{\mathrm{can,matched}}]$, total canonical frame shift $\delta_{\mathrm{renorm}}^{\text{C-m}} = H_{12} + D_{12}$, and sound-rescaled counterterm shift $\Delta_{\mathrm{ct}}^{\mathrm{sound}}\delta$ at fixed ratio $r = \beta_2/\beta_1 = 0.5$ **(c)** (`twobody_cont_grid.py`).

	Overtaking ($\epsilon_1 = +1, \epsilon_2 = +1$)				Head-on ($\epsilon_1 = +1, \epsilon_2 = -1$)			
(β_1, β_2)	$\delta_{0,b}^{\mathrm{ov}}$	$H_{\mathrm{ov}}^{\text{C-m}}$	$D_{\mathrm{ov}}^{\text{C-m}}$	$\delta_{\mathrm{renorm,ov}}^{\text{C-m}}$	$\delta_{0,b}^{\mathrm{ho}}$	$H_{\mathrm{ho}}^{\text{C-m}}$	$D_{\mathrm{ho}}^{\text{C-m}}$	$\delta_{\mathrm{renorm,ho}}^{\text{C-m}}$
$(0.10, 0.05)$	−0.016501	+0.006635	−0.956925	−0.950290	+0.020042	−0.009403	+1.588672	+1.579269
$(0.20, 0.10)$	−0.066270	+0.025369	−0.963038	−0.937670	+0.080671	−0.035425	+1.578916	+1.543491
$(0.30, 0.15)$	−0.150257	+0.052882	−0.973672	−0.920790	+0.183440	−0.072185	+1.559301	+1.487116
$(0.40, 0.20)$	−0.270127	+0.084691	−0.989524	−0.904833	+0.331026	−0.112842	+1.525697	+1.412854
$(0.60, 0.30)$	−0.631750	+0.145675	−1.041982	−0.896307	+0.785907	−0.179563	+1.404155	+1.224592
$\beta_1 \to 0$	0	0	−0.954930	−0.954930	0	0	+1.591549	+1.591549

6 Summary of Claims, Verification Scripts, and Failures Recorded

6.1 Summary Table of Claims and Verification Coverage

Claim / Formula	Label	Verification Script
Exact degree-2 factored bad-branch Hirota coefficient (4), classical phase (5), and canonical derivatives $-\partial\delta_{0,b}/\partial P_{0,j} = \Delta X'_j$ in all 4 sign sectors (6)	**(c)**	`twobody_cont_phase.py`, `twobody_cont_grid.py`
Small-amplitude expansions of $A_{12,b}$ and $\delta_{0,b}$ in both overtaking (7)–(8) and head-on (9)–(10) channels	**(c)**	`twobody_cont_phase.py`
Small-amplitude expansions of bad-branch determinant integrals $I_{12,b}, J_{12,b}$ in both channels (13)–(16), including $c^{\rm ov}_{2,\rm det}(r)$ and $c^{\rm ho}_{2,\rm det}(r) = -8r/\pi$	**(c)**	`twobody_cont_phase.py`
Lattice-matched one-body energy $F_{\rm matched}(\beta)$ (19), canonical momentum $p_{\rm can,matched}(\beta,\epsilon)$ (22), universal slope $c_1(p_{\rm can}) = -2C$, and Jacobian match (24)	**(c)**	`twobody_cont_phase.py`, `twobody_cont_det_antisym.py`
Small-amplitude frame shifts H_{12}, D_{12} under Prescription C-matched (25)–(28) ($D_{\rm ov} = -\frac{2}{\pi}(r^{-1} - r)$, $D_{\rm ho} = +\frac{2}{\pi}(r^{-1} + r)$) vs. Prescription M (30)–(32)	**(c)**	`twobody_cont_phase.py`
Direct on-shell sound-speed rescaling identities (33), Virial identity $5E_0 - 4vP_0 = \frac{2}{3}\sinh^3\beta$, and $\Delta^{\rm sound}_{\rm ct}\delta = O(\varepsilon^2) \to 0$ (34)–(36)	**(c)**	`twobody_cont_phase.py`
Exact finite-R bad-branch configuration-space kinetic norm (37) with finite constant $+\frac{v^2(\pi^2-6)}{9k^3}\dot{v}^2$	**(c)**	`twobody_cont_det_antisym.py`
Cutoff-endpoint / tadpole / higher-derivative sensitivity (39)–(40) and invariance of $C_{\rm matched}$ and $O(1)$ H_{12}, D_{12} under any analytic $O(\sinh^5\beta)$ counterterm	**(c)**	`twobody_cont_det_antisym.py`
Exact $r \leftrightarrow 1/r$ inversion $I_{\rm KdV}(r) + I_{\rm KdV}(1/r) = -\pi$ (41) and strict exchange antisymmetry (42) of $H_{12}, D_{12}, \Delta^{\rm sound}_{\rm ct}\delta$	**(c)**	`twobody_cont_phase.py`, `twobody_cont_det_antisym.py`
Large-Y continuum formula $\lim_{Y\to\infty}\mathcal{D}_0(Y) = -2C(15\sqrt{2} - 20)$ (44) and agreement of $C_{\rm matched} = \frac{1}{2\pi}$ ($\mathcal{D}_0 \to -0.386175$) with Toda data in Table 1	**(c)**	`twobody_cont_det_antisym.py`
Unsettled continuum status of (i) non-commuting projector Schur complement $\Delta L_{1,\rm Schur}$, (ii) Maslov/Levinson constant $Z = e^{i\delta_{\rm const}}$, and (iii) two-body collective momentum / Wronskian measure prescription selecting $\vartheta(1,2)$	**(d)**	`twobody_cont_det_antisym.py`
55-digit tabulation of all continuum phase components on r, (Y,r), (y_1,y_2), (β_1,β_2), and (α_1,α_2) grids	**(c)**	`twobody_cont_grid.py`

6.2 Failures Recorded

In compliance with `AGENTS.md`, we list every intermediate assertion failure encountered during the development of the verification scripts and how each was resolved without weakening any check:

1. **`twobody_cont_phase.py` (Series coefficients of $F_{\rm ac}$ and $F_{\rm matched}$):** Hand-estimated $O(\beta^7)$ coefficient of $F_{\rm ac}(\beta)$ (`-62/(945*pi)`) and $O(\beta^5)$ coefficient of $F_{\rm matched}(\beta) = F_{\rm ac}(\beta) - \sinh^3\beta/\pi$ (`-31/(45*pi)`, using $-\beta^5/\pi$ instead of $-\frac{1}{2}\beta^5/\pi$ from $\sinh^3\beta = \beta^3 + \frac{1}{2}\beta^5 + \frac{13}{120}\beta^7$)

failed the exact SymPy series assertions and were corrected to $+\frac{97}{1890\pi}\beta^7$ for $F_{\rm ac}$ and $-\frac{17}{90\pi}\beta^5 - \frac{431}{7560\pi}\beta^7$ for $F_{\rm matched}$ (giving $p_{\rm can,matched} = -\frac{\epsilon}{\pi}\beta + \frac{10\epsilon}{9\pi}\beta^3 + \frac{2\epsilon}{9\pi}\beta^5$).

2. **`twobody_cont_phase.py` (Subleading Jacobian term):** The $O(\beta^2)$ term of $\beta^2 \frac{\mathrm{d}p_{\rm can,matched}/\mathrm{d}\beta}{\mathrm{d}P_0/\mathrm{d}\beta}$ was briefly changed from $+\frac{5}{2\pi}\beta^2$ to $+\frac{2}{\pi}\beta^2$ and failed the SymPy assertion at line 203; restored to $+\frac{5}{2\pi}\beta^2$.

3. **`twobody_cont_phase.py` (Bad-branch $A_{\rm ov}/A_0(r)$ subleading ratio):** Before factoring out $w_j^2 - k_j^2 - k_j^4 = 0$, initial test targets $1 \mp \frac{2}{3}r\varepsilon^2$ and $1 + \frac{13}{9}r\varepsilon^2$ for $A_{\rm ov}/A_0(r)$ failed; factoring $A_{12,b}$ into (4) both reduced the script runtime from $\sim 200\,$s to $20\,$s and proved $A_{\rm ov}/A_0(r) = 1 + r\varepsilon^2 + \frac{1}{2}r^2\varepsilon^4 + O(\varepsilon^6)$.

4. **`twobody_cont_phase.py` (Exponent of r in $-\gamma_2 M_1$ for $D_{\rm mode}$):** In `D_ov_mode_lead` and `D_ho_mode_lead`, $-\gamma_2 M_1 = +\frac{2}{r^2\varepsilon}$ was initially typed as `2/(r_s*eps_s)` instead of `2/(r_s**2*eps_s)`; caught by SymPy expansion assertions and fixed.

5. **`twobody_cont_phase.py` (Power of r in $4P_{0,2}\Delta X_2'$ for $\Delta_{\rm ct}^{\rm sound}\delta^{\rm ho}$):** In `c2_sound_ho`, $4P_{0,2}\Delta X_2'$ was initially written as $-\frac{16}{3}r^2\varepsilon^2$ instead of $-\frac{16}{3}r\varepsilon^2$; caught by the SymPy assertion and corrected to $\Delta_{\rm ct}^{\rm sound}\delta^{\rm ho} = -\frac{2}{\pi}r\varepsilon^2 + O(\varepsilon^4)$, consistent with Euler's homogeneous function theorem for $\delta_{0,\rm ho}^{(0)} = 4\beta_1\beta_2 \propto |P_{0,1}P_{0,2}|^{1/3}$.

6. **`twobody_cont_det_antisym.py` (Sign of finite kinetic-norm integral):** In the initial draft, $\int_{-\infty}^{\infty}[(\tanh z + z\,\mathrm{sech}^2 z)^2 - 1]\,\mathrm{d}z$ was written as $-\frac{2(\pi^2-6)}{9}$ instead of $+\frac{\pi^2-6}{9}$ (since $\int_0^\infty(-\mathrm{sech}^2 z + 2z\tanh z\,\mathrm{sech}^2 z)\,\mathrm{d}z = -1+1 = 0$, leaving $2\int_0^\infty z^2\,\mathrm{sech}^4 z\,\mathrm{d}z = +\frac{\pi^2-6}{9} > 0$). Caught by the 70-digit `mpmath` quadrature assertion at line 61 and corrected.

7. **`twobody_cont_det_antisym.py` ($O(\beta^2)$ coefficient of $F_{\rm matched}/P_0$):** Line 102 initially carried the un-updated coefficient $-\frac{79}{60\pi}$ instead of $-\frac{47}{60\pi}$ from $(1-\frac{17}{30}\beta^2)/(1+\beta^2) = 1-\frac{47}{30}\beta^2$; caught by SymPy assertion and corrected.

8. **`twobody_cont_det_antisym.py` (Prefactor of $\lim_{Y\to\infty}\mathcal{D}_0(Y)$ in terms of C):** Line 179 initially wrote $-8C(15\sqrt{2}-20)$ instead of $-2C(15\sqrt{2}-20)$ (since $8\cdot[-C(15\sqrt{2}-20)/4] = -2C(15\sqrt{2}-20)$, which equals $-\frac{15\sqrt{2}-20}{\pi}$ at $C_{\rm matched} = \frac{1}{2\pi}$). Caught by SymPy limit assertion and corrected.

9. **`twobody_cont_grid.py` (Derivative $\mathrm{d}I_{\rm KdV}/\mathrm{d}r$, closed-form $\delta_{0,b}$, and good-branch resonance boundary):** (a) Line 260 initially included a non-cancelling boundary term in $\mathrm{d}I_{\rm KdV}/\mathrm{d}r$, caught by 55-digit `mp.diff` check and simplified to $\frac{4\ln r}{\pi(1-r^2)}$; (b) `delta0_bad_mp` and `dP0_db_bad_mp` initially had transcription typos caught by the 50-digit path-quadrature check on $[0.1, 0.2]$ and corrected to (5) and $\frac{2}{3}\sinh\beta\sinh 3\beta$; (c) the initial good-branch pairs $(0.8, 0.4)$ and $(0.6, 0.45)$ had $\alpha_1+\alpha_2 > \pi/3$ where $A_{12,g}^{\rm ho} < 0$, and were replaced with regular-chamber pairs satisfying $\alpha_1+\alpha_2 < \pi/3$; (d) the finite-amplitude convergence check at $\varepsilon = 0.1, 0.2$ was upgraded to subtract the exact $O(\varepsilon^2)$ term and verify $O(\varepsilon^4)$ scaling (ratio ≈ 16).

References

[1] L. D. Faddeev and V. E. Korepin, "Quantum theory of solitons," *Phys. Rept.* **42** (1978) 1–87 (full text verified in project literature).

[2] E. Corrigan and C. Zambon, "Integrable defects in the Boussinesq equations," arXiv:2306.03772 (full text verified in project literature; dictionary $u_{\rm CZ} = -3\phi$, $L_{\rm CZ} = 9L$).

[3] Internal repository packages and audits (all full texts read and cross-checked): phase1/review.tex, phase2/report.tex, phase2/errata.tex, phase2/oneloop_phase.tex, phase2/det_projected_oneloop.tex, phase2/audit_det_projected.tex, phase2/audit_determinant_foundations.tex, phase2/onebody_scheme.tex, phase2/audit_onebody_scheme.tex, phase2/onebody_dispersion_test.tex, phase2/toda_route.tex, phase2/smx_candidates.tex, phase2/audit_smx.tex, phase2/frame_dictionary.tex, phase2/audit_frame_dictionary.tex.